

LAB211 Assignment

Type:
Code:
LOC:
Slot(s):

Short Assignment
J1.S.P0070
150
3

Title

Login system of the Tien Phong Bank's Ebank

Background

(Module extracted from TienPhong Bank, ebank project)

Program Specifications

Login function of Ebank system includes:

1. Vietnamese
2. English
3. Exit

If the user selects 1: Switch the interface language to Vietnamese and perform "check login" function.

If the user selects 2: Keep the English interface and perform "check login" function.

Check the Login function:

1. Require to input an account number :
 - Check the account number must be a number
 - Each account number must be 10 digits.
2. Input a password:
 - Check length of the password which is enough length or not
 - Check the password which includes alphanumeric or not.
3. Require to input one character or multiple characters in the captcha:
 - Create a random captcha code in each login.
 - Check the input characters of captcha are correct or not.

Function details:

Function 1: Display a menu and ask users to select an option.

- User runs the program. The program prompts users to select an option.
- User selects an option, perform **Function 2**.

Function 2: Perform language based on the selected option.

Function 1: Change language

- Use the resourceBundle to get the value of the language according to the key in 2 files as En.properties and Vi.properties which are corresponding English and Vietnamese

.Function 2: Checking account number

- Using Regular Expression to check the validity of the account number was input at the request of the assignment.
- If the account number is invalid, it returns an error message corresponding to the language you chose.

Function 3: Check the password.

- Use a RegularExpression to check the validation of the inputted password.
- If the password is invalid, then return an error message corresponding **to the language** you have just selected.

Function 4: Generate a random captcha code.

- Use the Random function to generate a random sequence, and then converter to characters (type char)
- .

Function 5: Check the captcha code

- Use the function contains () to check the captcha characters input exists in the captcha string was born at first did not.
- If the captcha characters are invalid, then returns the error message that corresponds **to the language** you have selected.

Function 6: Login

- Allows the user to input the account number, password and captcha from the keyboard.
- If a valid account number, then allows the user to input a password and if the account number entered is not valid, then print out the error screen and allows the user to input again.
- If valid password then allows users to input captcha. Otherwise then print out the error screen and allows the user to input again.
- If the captcha is not valid then print out the error screen.

EExpectation of User interface:

Student must implement methods

setLocate
checkAccountNumber
checkPassword

in startup code.

- Create a Main class to display the selected language in the Menu.
- Create the Ebank class including the following functions:

Function 1: Convert the languages.

- The program will switches from this language to another one, so that the result is the elements is defined in the Language.properties file will convert to another one.
- Implement the function: void setLocate(Locate locate).
 - Input :
 - locate: name of properties file which needs to be converted.

Function 2: Check the account number.

- The program checks an account number, so that it must be one number as the defined requirements.
- Implement the function: String checkAccountNumber (String accountNumber)
 - Input :
 - accountNumber: account number needs to be checked.
 - Return value: messages about the value of the account number.

Function 3: Check the password.

- The program checks the password, so that it must assure the defined requirements.
- Implement the function: checkPassword String (String password)
 - Input :
 - password: the password needs to be checked .
 - The value returns: messages about the value of the password.

Functions 4: Generate a random captcha code.

- The program generates a random captcha code, so that the it must assure the defined requirements.
- Implement the function: String generateCaptcha ()
 - Return value: a random captcha string.

Function 5: Check the captcha code.

- The program checks the captcha code, so that the result is a captcha string which must assure the defined requirements.
 - Implement the function: String checkCaptcha (String captchaInput, String captchaGenerate)
 - Input :

- captchaInput: Captcha is inputted
- captchaGenerate: Captcha is generated
- The value returns: The message about the value of the captcha.